American Express Customer Attrition Analysis

Group 21

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Introduction

This report analyzes customer attrition for American Express using a credit card customer dataset that contains records for over 10,000 customers. The dataset includes information on customer demographics, account characteristics, credit utilization, and transaction behavior across multiple time periods. Following data cleaning and the removal of observations with missing values, the final dataset was used to examine patterns in customer engagement and attrition.

The purpose of this analysis is to determine which factors are most strongly associated with customer attrition and to identify measurable indicators that can be used to predict attrition risk. Tableau was used to visually explore relationships within the data, while regression analysis was applied to quantify the impact of key variables on the probability of attrition. Together, these methods support data-driven decision-making and provide insight into customer behavior.

The analysis shows that customer engagement is the primary driver of attrition. Customers with lower transaction counts, lower transaction amounts, and declining activity between Q1 and Q4 are significantly more likely to attrite, while customers who consistently use their cards and maintain revolving balances are more likely to remain active. Based on these findings, this report recommends implementing targeted, usage-based incentives and early intervention strategies focused on customers showing early signs of declining engagement in order to reduce attrition and improve long-term customer retention.

Data Exploration with Tableau

You can explore the full interactive Tableau dashboard by clicking here, where the visualizations break down the key factors that influence customer attrition.

Data Overview and Statistical Analysis

Data Overview

The dataset contained credit card information for over 10,000 customers. Initially, data were available for 22 different variables; however, the final model includes 13 variables: months inactive, total revolving balance, total transaction amount, total transaction count, utilization ratio, relationship count, card type, gender, education level, marital status, income category, change in transaction amount between Q1 and Q4, and change in transaction count between Q1 and Q4. Some observations contained missing data, which required removal from the analysis. Specifically, 287 customers were missing revolving balance values, 97 were missing transaction amount data, and 4 were missing gender information. Because regression analysis cannot be performed with missing values, these observations were removed from the dataset.

Attempted Models

We estimated 15 different regression models to determine which variables best explained the probability of customer attrition. For each model, we evaluated adjusted R^2, standard error, and individual p-values to assess model performance and variable significance. During this process, several variables exhibited non-linear patterns, requiring transformation. Total revolving balance and total transaction amount were transformed using natural logarithms, while transaction count was transformed using a square root transformation due to its exponential pattern. These adjustments improved model fit, increased adjusted R^2, and reduced standard errors.

After testing multiple model specifications, some variables, such as credit limit, were found to have little impact on customer attrition and were excluded from the final model. This result was unexpected, as higher credit limits were initially believed to influence card usage behavior. After evaluating all results, we selected the final regression model based on its overall fit and predictive performance.

Equations

Final Population Regression Equation:

$$\begin{aligned} Attrtion(d) = &-0.06 + 0.04(Months\ Inactive) - 0.03\ln(Total\ \ Resolving\ Balance) \\ &+ 0.29\ln(Total\ Transaction\ Amount) - 0.20\sqrt{(Total\ Tranaction\ Count)} \\ &+ 0.02(Average\ Utilization\ Ratio) - 0.04(Relationship\ Count) \\ &+ 0.05\big(Gold(d)\big) + 0.11\big(Platinum(d)\big) - 0.05\big(Male(d)\big) + 0.04\big(PhD(d)\big) \\ &- 0.04\big(Married(d)\big) + 0.03\big(\$120k + (d)\big) \\ &- 0.07(Total\ Amount\ Change\ Q1 - Q4) \\ &- 0.32(Total\ Count\ Change\ Q1 - Q4) + \varepsilon \end{aligned}$$

Estimated Sample Regression Equation:

$$\begin{aligned} \widehat{Attrition}(d) = &-0.06 + 0.04(Months\ Inactive) - 0.03\ln(Total\ \ Resolving\ Balance) \\ &+ 0.29\ln(Total\ Transaction\ Amount) - 0.20\sqrt{(Total\ Tranaction\ Count)} \\ &+ 0.02(Average\ Utilization\ Ratio) - 0.04(Relationship\ Count) \\ &+ 0.05\big(Gold(d)\big) + 0.11\big(Platinum(d)\big) - 0.05\big(Male(d)\big) + 0.04\big(PhD(d)\big) \\ &- 0.04\big(Married(d)\big) + 0.03\big(\$120k + (d)\big) \\ &- 0.07(Total\ Amount\ Change\ Q1 - Q4) \\ &- 0.32(Total\ Count\ Change\ Q1 - Q4) \end{aligned}$$

Fit of the Model

Adjusted R^2 Interpretation: We are 38.9% of the way towards perfectly predicting if a customer has attrited or not using this model.

Standard Error: Predictions using this model are off by an average of 0.2875 percentage points, on average all else constant.

The full regression output for the final model is provided in **Figure 1** in the Appendix.

Test of Individual Significance: Because the p-value for each of the variables besides the Platinum(d) had a p-value of less than 0.10 We can reject the null and prove that each of the variables below were significant.

We left Platinum(d) in the final model as it missed the cut off by 0.01, also it goes with the other dummy variables of card type. The removal of this variable led to a decrease in adjusted R^2 so we decided to leave it in.

Coefficient Interpretations

Months Inactive: As months inactive in the past 12 months increases by 1 month, the probability of attrition increases by 4 percentage points, on average and all else constant.

Ln(Total Resolving Balance): As total revolving balance increases by 1 percent, the probability of attrition decreases by about 0.03 percentage points, on average and all else constant.

Ln(Total Transaction Amount): As total transaction amount increases by 1 percent, the probability of attrition increases by about 0.29 percentage points, on average and all else constant.

$\sqrt{Transaction\ Count}$: As the square root of transaction count increases by 1 unit, the probability of attrition decreases by about 20 percentage points, on average and all else constant.

Average Utilization Ratio: As average utilization ratio increases by 1 unit, the probability of attrition increases by about 2 percentage points, on average and all else constant.

Gold Card (dummy): Customers with a Gold card are about 5 percentage points more likely to attrite than customers in the Blue card category, on average and all else constant.

Platinum Card (dummy): Customers with a Platinum card are about 11 percentage points more likely to attrite than customers in the Blue card category, on average and all else constant.

Male (dummy): Male customers are about 5 percentage points less likely to attrite than female customers, on average and all else constant.

PhD (dummy): Customers with a PhD are about 4 percentage points more likely to attrite than customers with all other types of education, on average and all else constant.

Income $120+ (dummy): Customers earning $120,000 or more are about 3 percentage points more likely to attrite than customers who don't, on average and all else constant.

Change Between Q1 and Q4 (Total Transaction Amount): As the total transaction amount change ratio from Q4 to Q1 increases by 1, the probability of attrition decreases by about 7 percentage points, on average and all else constant.

Change Between Q1 and Q4 (Total Transaction Count): As the total transaction count change ratio from Q4 to Q1 increases by 1, the probability of attrition decreases by about 32 percentage points, on average and all else constant.

Common Violations

Linearity: Non-linear relationships were detected. This was evident from residual plots and from the improved model fit after transforming several variables. The residual plots from the final model are provided in **Figure 3** in the Appendix. Non-linearity can cause biased predictions and poor models fit if left unaddressed. This violation was fixed by applying logarithmic and square-root transformations to the respective variables, which improved the adjusted R^2 and better captured the underlying relationships.

Multicollinearity: Multicollinearity was detected between total transaction amount and total transaction count, as these variables are highly related. Removing one of these variables reduced the adjusted R^2 and weakened the model's predictive performance. Because the goal of the model was prediction, both variables were kept in the model despite the presence of multicollinearity.

A correlation matrix used to assess multicollinearity among the independent variables is shown in **Figure 2** in the Appendix.

Heteroskedasticity: Some evidence of heteroskedasticity was observed through residual plots, with slightly increasing variance at higher predicted probabilities. However, the pattern was not abnormal and verry slim, the large sample size reduces the practical impact of this issue. As a result, no correction was applied.

Endogeneity: No evidence of endogeneity was detected in the model. The independent variables were determined prior to the attrition outcome, and no clear omitted variables or reverse causality issues were identified. Endogeneity can lead to biased and inconsistent coefficient estimates, but because it was not present, no corrective action was necessary.

Relevance of Findings

We found several trends that will be important for the team at American Express. Customer Engagement is the strongest predictor of attrition. Clients with low transaction count and transaction amounts are specifically more likely to become inactive. Customers who use

their cards consistently are more likely to continue spending time with the company. This can be helpful in many ways and suggests that frequency use is much more important than simply just holding an account.

The regression output showed us that another ley sign of customer attrition is the decline between activity in Q1 and Q4. Customers who complete fewer transactions and show declining activity over time are significantly more likely to attrite. Thews findings support that declining engagement is a measurable and early warning sign of a customer about to attrite. Credit utilization and resolving balance is also a significant indicator, customers who use up their credit on a consistent basis are more likely to stay with the company, similar customers with consistent balances are also likely to stay and unlikely to attrite.

Some surprising from the regression output was the fact that credit limit ended up not being significant. This surprised us as we would suspect that the higher credit limit you have on that card the more you would be spending on it and the more activity you would have, but we found that was not the case. We also found that categories like marital status had an impact on whether you attrite or not and found that married couples attrite less often than single or divorces customers. We were also surprised to find out that age did not play a role in the chance of attriting or not, this was surprising because we assumed that younger customers who are just starting out are more likely to leave AMEX but that was not the case.

Recommendations

Tableau

Based on the insights we gained from the tableau, we recommend launching usage-based incentives, such as bonus points for completing a certain number of monthly transactions. We also recommend offering target-based rewards aimed at customers with low or inconsistent spending patterns that will get them using their card more. Finally, we suggest sending reminders to customers with low activity to get them into categories with lower attrition percentages.

Regression Output

When looking at the Regression Output, you can see that customer behavior such as inactivity, transaction patterns, and relationship depth are vital predictors of attrition. We recommend for American Express to keep a consistent amount of spending instead of sporadic volume spending. High transaction amounts do reduce attrition risk, but it does not mean that it will be retained. Spending on an everyday basis is the best way to increase customer loyalty and keep risks at a low. Based on the regression results, we recommend using the model as an early-warning system to identify customers whose engagement is

beginning to decline, particularly those showing increases in months inactive and decreases in transaction count or amount between Q1 and Q4. Customers flagged as high risk by the model should be prioritized for proactive outreach before full attrition occurs. In addition, the regression can be used to segment customers by attrition risk level, allowing American Express to focus retention efforts on the customers most likely to leave while maintaining efficient use of resources.

Conclusion

This analysis shows that customer attrition at American Express is driven primarily by how actively customers use their cards. Customers who complete fewer transactions, spend less overall, or show a noticeable drop-in activity from Q1 to Q4 are much more likely to disengage. On the other hand, customers who use their cards consistently and maintain ongoing balances tend to remain with the company. These patterns suggest that shifts in usage behavior provide clear and early signals of attrition risk.

Looking ahead, these insights create an opportunity for American Express to intervene before customers fully disengage. Focusing retention efforts on customers whose activity is beginning to decline, rather than waiting until inactivity occurs, can help reduce attrition and improve long-term customer relationships. Usage-based incentives, personalized outreach, and targeted reminders are practical strategies that align directly with the behaviors identified in this analysis. In addition, the regression model developed in this project can serve as a predictive tool to help prioritize high-risk customers and guide data-driven retention decisions.

For any questions or follow-up regarding this analysis, the project team can be contacted at liamparker@arizona.edu.

Appendix

Figure 1 - Regression Output

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.624132411
R Square	0.389541267
Adjusted R Square	0.388662367
Standard Error	0.287455748
Observations	9739

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	14	512.7245431	36.62318165	443.2146182	0
Residual	9724	803.5019689	0.082630807		
Total	9738	1316.226512			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-0.059504276	0.060849148	-0.97789827	0.328148967	-0.178781261	0.059772708
Months Inactive 12 mo	0.040047036	0.002891941	13.8478041	3.36743E-43	0.03437823	0.045715842
(Ln) tot resolving bal	-0.030450948	0.001207376	-25.2207578	5.2313E-136	-0.032817656	-0.028084239
Ln(Tot Trans Amt)	0.289418168	0.010739749	26.94831743	2.5257E-154	0.268366026	0.31047031
SQRT(Trans Cnt)	-0.204346556	0.004545595	-44.9548573	0	-0.213256866	-0.195436245
Avg Utilization Ratio	0.02414308	0.014307515	1.687440495	0.091550754	-0.003902625	0.052188786
Relationship Cnt	-0.037516818	0.001991322	-18.8401539	8.51384E-78	-0.041420223	-0.033613412
Gold (d)	0.053801944	0.028006651	1.921041653	0.05475566	-0.001096918	0.108700806
Platinum (d)	0.107145555	0.068015178	1.575318294	0.115215525	-0.02617834	0.24046945
Male (d)	-0.049833011	0.006446031	-7.73080557	1.1746E-14	-0.062468573	-0.03719745
PhD(d)	0.036954128	0.014206802	2.60115738	0.009305053	0.009105841	0.064802415
Married(d)	-0.044045359	0.005916446	-7.44456361	1.05523E-13	-0.055642823	-0.032447894
$120K +	0.027389319	0.01194266	2.293401851	0.021846113	0.003979221	0.050799418
Change Between Q1 ar	-0.067931738	0.014543556	-4.67091682	3.0388E-06	-0.096440132	-0.039423345
Total Count Chane Bet	-0.315663691	0.013331923	-23.677281	1.5247E-120	-0.341797033	-0.289530349

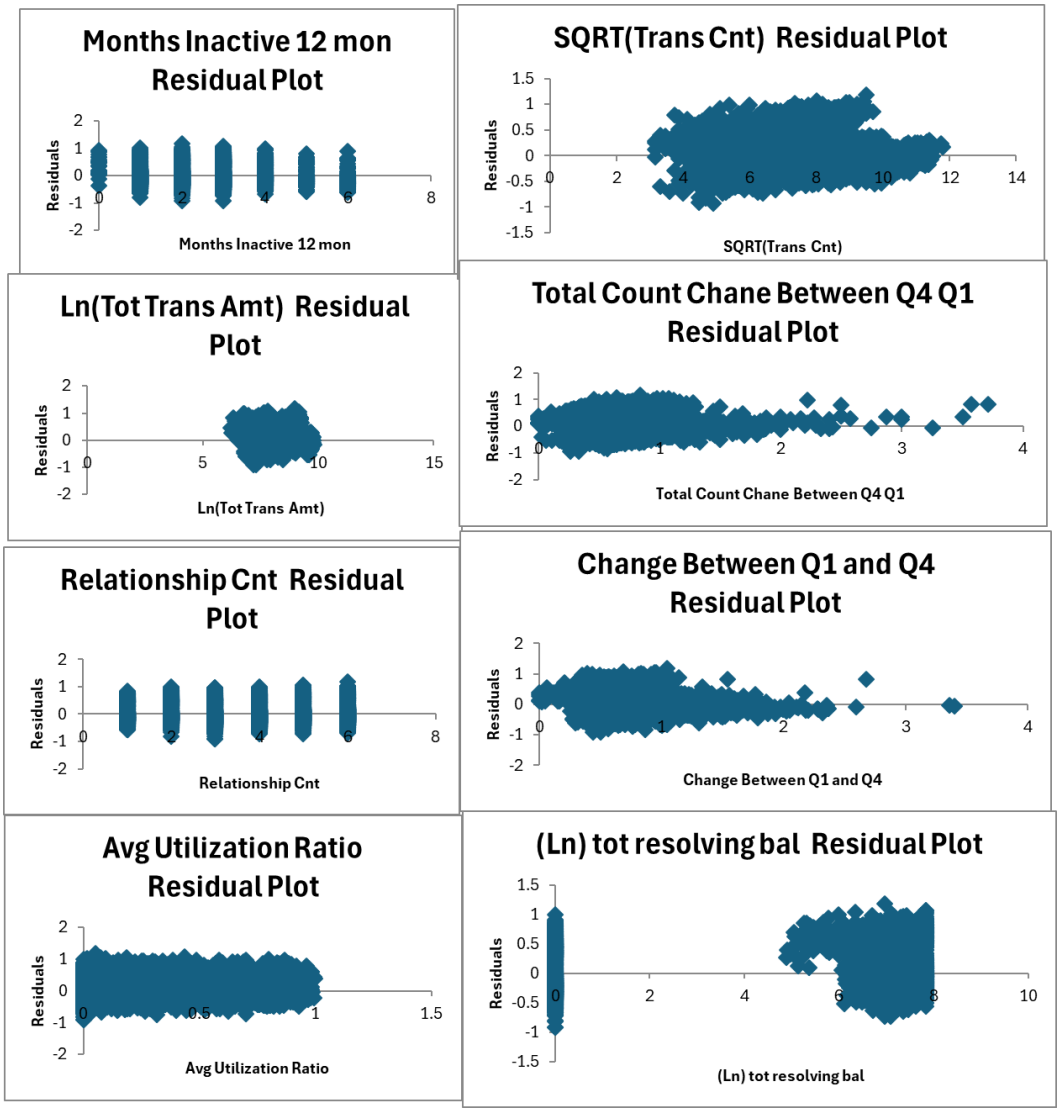
Return to Fit of the Model

Figure 2 - Correlation Matrix

	Residuals	Months Inactive 12 mon	(Ln) tot resolving bal	n(Tot Trans Amt)	SQRT(Trans Cnt)	Avg Utilization Ratio	Relationship Cnt	Gold (d)	Platinum (d)	Male (d)	PhD(d)	Married(d)	$120K +	Between Q1	unt Chane Betwee
Residuals	1														
Months Inactive 12	1.93516E-16	1													
(Ln) tot resolving b	-8.49885E-15	-0.053256	1												
Ln(Tot Trans Amt)	-6.12891E-15	-0.032673	0.054828	1											
SQRT(Trans Cnt)	-6.96279E-15	-0.042061	0.076767	0.901349	1										
Avg Utilization Rati	-4.87191E-15	-0.008728	0.595538	-0.033495	0.009845	1									
Relationship Cnt	4.01072E-15	-0.004895	0.028680	-0.297592	-0.215191	0.066986	1								
Gold (d)	-5.46636E-16	-0.000794	0.018824	0.090968	0.068153	-0.083836	-0.052599	1							
Platinum (d)	-4.54456E-16	-0.009653	0.006063	0.042803	0.032303	-0.036412	-0.040995	-0.004578	1						
Male (d)	1.98517E-16	-0.012603	0.029509	-0.053905	-0.079762	-0.257554	0.004280	0.038357	0.002465	1					
PhD(d)	-1.94645E-17	0.005484	-0.007944	-0.006743	-0.005142	-0.010468	-0.016879	-0.003812	0.014063	-0.019278	1				
Married(d)	-4.76601E-17	-0.008328	0.033416	-0.119214	-0.136503	0.047255	0.020187	-0.020399	-0.015949	0.012924	-0.000445	1			
$120K +	8.38631E-17	-0.005315	0.010643	-0.016057	-0.020353	-0.151125	-0.004080	0.035227	0.016042	0.292636	0.004992	0.017135	1		
Change Between Q	-6.40217E-17	-0.029533	0.067135	0.048629	0.002689	0.034845	0.049755	0.006762	0.000594	0.025879	-0.018042	0.052985	-0.004598	1	
Total Count Chane	-2.53262E-16	-0.036818	0.094516	0.109484	0.105193	0.074761	0.043573	-0.003923	-0.008574	-0.004269	-0.017510	0.014804	-0.027241	0.385934	1

Return to Multicollinearity

Figure 3 – Residual Plots



Return to Linearity